

ASSIGNMENT 3

IDEA: dinosaur game - inspired but with fish?

3 CLASSES:

1. GAME
2. FISHING LINE WITH BAIT
3. FISH

ARRAY WITH MIN. OF 10 ELEMENTS:

- values of distance between each bait

INPUTS:

- UP & DOWN ARROWS
- SPACE BAR (IF JUMPING)

MATH:

- SPEED AS TIME PASSES
- POSITIONS AND SIZE OF BAIT & FISH

★ COLLISION DETECTION!!!

NOTES:

- USE WHILE LOOPS FOR BAIT
 - BAIT MOVES; FISH STATIC
 - CREATE SCORE COUNTER (LOW PRIORITY)
 - IF STATEMENTS FOR WIN/LOSE SCREEN ✓
 - IF FISH HITS BAIT OR FISHING LINE, GAME OVER!!
- (MAYBE) ↗

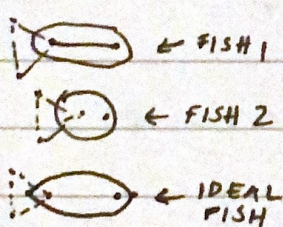
PROBLEMS:

1. keeping drawings together
2. figure out bait movement and make array for positions
3. figure out how to make game restart
4. collision
5. what happens when fish hits bait
6. bait comes from left side

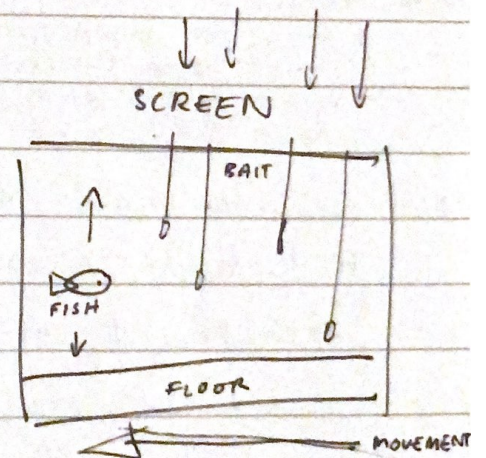
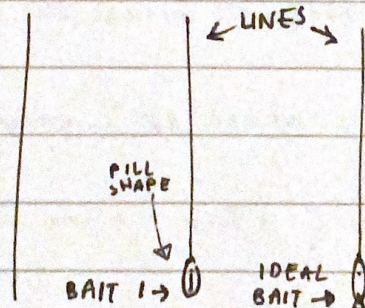
SOLUTIONS:

1. use one/main shape's co-ordinates and add/subtract from x & y positions
2. make variable for x-position y-position stays the same
3. look at previous class demo
4. module (review)
5. take away heart? (if condition)
6. pos.x + 800, then move x amount of pixels left (subtract from pos)

DRAWINGS:



(2 SHAPES)



(CIRCLE/CIRCLE
POINT RECTANGLE
RECTANGLE-RECTANGLE)

- ~~THE~~ HARD TO FIGURE OUT ANY OF THE THREE TYPES OF COLLISION FOR FISH & BAIT

PROBLEM LOG

COLLISION HITBOX:

- RECTANGLES ARE IN DIFFERENT CLASSES (FISH & FISHBAIT)
- INPUT IS NOT A MOUSE, MUST USE FISH'S X POSITION
- FISH'S X POSITION USES THIS.POS AND VARIABLE CAN'T BE CALLED IN FISHBAIT CLASS (?)
- FISHBAIT CLASS ALSO USES THIS.POS
- IF COLLISION IS MADE IN ANOTHER CLASS, BOTH FISH & FISHBAIT'S POSITIONS MUST BE KNOWN BUT FISH Y IS CONSTANTLY CHANGING AND FISHBAIT X & Y ARE ALL DIFFERENT
- RECTANGLES DRAWN FOR FISH & BAIT ARE KEPT IN ~~THEIR~~ THE DRAW ____ {} Voids (CAN'T COMBINE???)

RESTART:

- CAN'T USE SETUP(); FUNCTION LIKE PREVIOUS DEMO

FISH BAIT LINE:

- CALCULATE DISTANCE FROM CENTRE OF BAIT TO TOP OF SCREEN?
- X - USE BAIT ~~W~~ Y CO-ORDINATE AND $X=0$ (TOP OF SCREEN)
- ✓ - DRAW LINE TO BE HEIGHT OF SCREEN AT ALL TIMES

BAIT

~~FROM~~ POSITION:

- WHEN USING POS.X, BAIT COMES FROM LEFT SIDE
- STILL DOESN'T APPEAR AFTER USING $POS.X = -POS.X$ (TO FLIP DIRECTION)
- MUST CHANGE STARTING X POSITION AND DIRECTION (TO LEFT)

FISH SHAPE:

- 4 ARCS TOO COMPLICATED TO DRAW FOR FISH & BAIT
- JUST USE ELLIPSE
- POINT (DRAWN WITH DRAW-LINE) DOESN'T WORK FOR EYE
- JUST USE CIRCLE

★ IF COLLISION WORKS, BAIT COUNTER WOULD BE SOMETHING LIKE:

```
if (collision math) & to check  
{  
  DRAW.Text ("score"); ← score = (math & something like i++)  
}
```

- JUST CALCULATE SCORE BEFORE PRINTING ON SCREEN

FISH GAME

